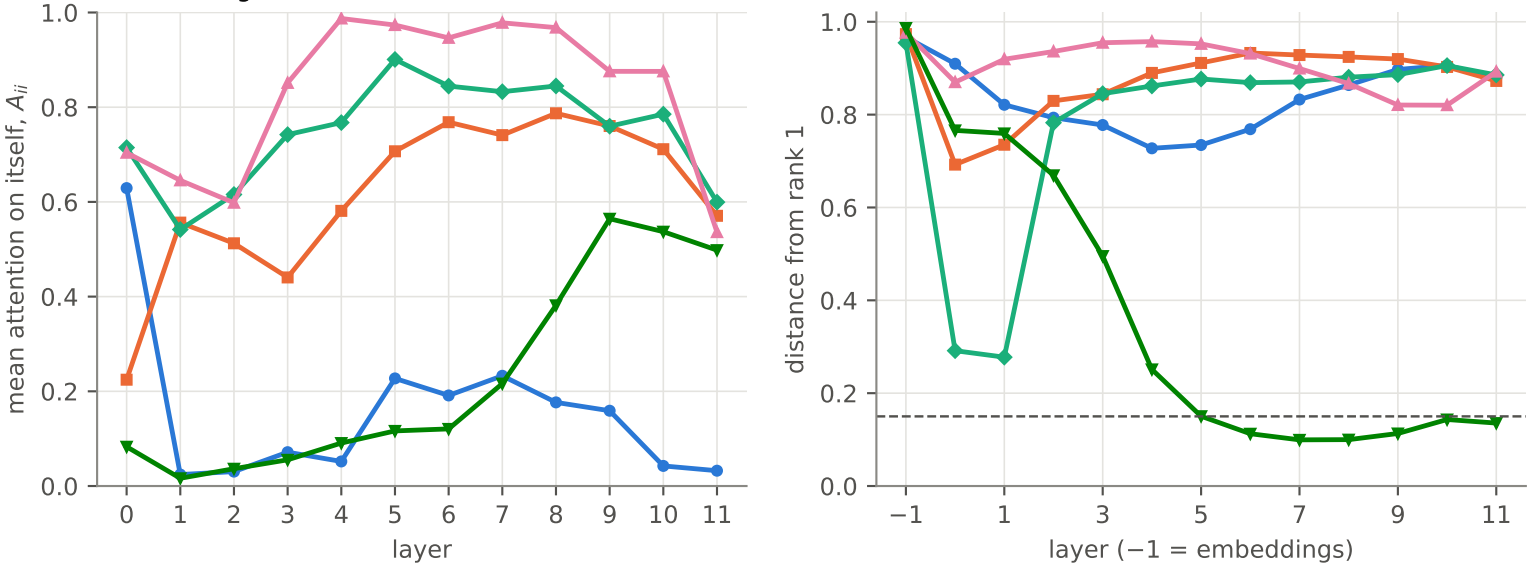
(a) self-weight (bias is 0 in all)
mean attention on itself, A_{ii}
1.0
0.8
0.6
0.4
0.2
0.0
0 1 2 3 4 5 6 7 8 9 10 11
layer
(b) rank-1 distance of the residual stream
distance from rank 1
1.0
0.8
0.6
0.4
0.2
0.0
−1 1 3 5 7 9 11
layer (−1 = embeddings)
baseline (GPT-2)
three_residual
approach_1 (step 3000)
approach_2
approach_4